

Tiger Lu

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APPOINTMENTS	Center for Computational Astrophysics , Flatiron Institute, New York, NY Flatiron Research Fellow	Sep 2025 –
	Yale University , New Haven, CT Graduate Researcher	Aug 2020 – May 2025
	University of California, Berkeley , Berkeley, CA Summer Undergraduate Researcher	May 2019 – Aug 2019
	Carnegie Observatories , Pasadena, CA Summer Undergraduate Researcher	May 2018 – Aug 2018

EDUCATION	Yale University , New Haven, Connecticut, USA	
	▪ Ph.D. in Astronomy with Distinction	Aug 2020 – May 2025
	• Thesis: Planets are not Points: the Profound Effect of Planetary Structure on Exoplanet System Architectures	
	• Advisers: Prof. Gregory Laughlin, Prof. Malena Rice	
	▪ M.S., M.Phil. in Astronomy	Aug 2020 – Aug 2022
	California Institute of Technology , Pasadena, California, USA	
	▪ B.S. in Astrophysics	Aug 2016 – Jun 2020
	▪ Minor in Computer Science	

PUBLICATIONS 226 citations. h -index = 9.
Italics: Student directly supervised by T. Lu

FIRST-AUTHOR

- [9] **Lu, T.**, Hernandez, D.M., Rice, M. and Su, Y. In review.
- [8] **Lu, T.** and Brown, Garrett. [A Coordinate System for Dynamical Instabilities in Hierarchical Systems in REBOUND](#). In review.
- [7] **Lu, T.**, Millholland, S.C., Rice, M., Black, B., Bardalez Gagliuffi, D.C., Balmer, W.O., Pueyo, L., Giovinazzi, M.R. and Brandt, T.D. [The Architecture of the 14 Herculis System Suggests Primordial Ejection of a Massive Planet](#). *ApJ* **1006**, 159
- [6] **Lu, T.**, Tajer, H., Hernandez, D.M., Rein, H., *Liu, Y.* and Rice, M., 2025. [Collisional Fragmentation Support in TRACE](#). *RNAAS* **9**, 5
- [5] **Lu, T.**, Li, G., Cassese, B. and Lin, D.N.C., 2025. [The Dynamical History of HIP-41378 f - Oblique Exorings Masquerading as a Puffy Planet](#). *ApJ* **980**, 39
- [4] **Lu, T.**, An, Q., Li, G., Millholland, S.C., Rice, M., Brandt, G.M. and Brandt, T.D., 2025. [Planet-Planet Scattering and ZLK Migration – the Dynamical History of HAT-P-11](#). *ApJ* **979**, 218
- [3] **Lu, T.**, Hernandez, D.M. and Rein, H., 2024. [TRACE: a Time-Reversible Algorithm for Close Encounters](#). *MNRAS* **533**, 3708
- [2] **Lu, T.**, Rein, H., Tamayo, D., Hadden, S., Mardling, R., Millholland, S.C. and Laughlin, G., 2023. [Self-Consistent Spin, Tidal and Dynamical Equations of Motion in the REBOUNDx Framework](#). *ApJ* **948**, 41
- [1] **Lu, T.** and Laughlin, G., 2022. [Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9](#). *PSJ* **3**, 221

STUDENT-LEAD

- [4] *Yang, E.*, **Lu, T.**, Yahalomi, D.A. and Winn, J.N., 2026. [A Massive Hot-Jupiter Companion that Disfavors Giant Planet Formation Beyond the Water-Ice Line](#). *ApJL* **1004**, L29
- [3] *Im, H.*, **Lu, T.**, Rice, M., Tran, Q., Li, G. and Naoz, S., 2026. [Observational and Dynamical Constraints on an Unseen Outer Perturber in the GJ 436 Hot Neptune System](#). *ApJ* **1003**, 84

- [2] *Liu, Y., Lu, T. and Rice, M., 2026. Double Hot Jupiter Formation through Mirrored ZLK Migration in Binary Star Systems: The Case of WASP-94. ApJL 997, L41*
- [1] *Liu, Y., Lu, T. and Rice, M., 2025. The Formation of Double Hot Jupiter Systems Through ZLK Migration. ApJ 986, 103*

SIGNIFICANT CONTRIBUTION

- [6] Rescigno, F., Stalport, M., Anna John, A., **Lu, T.**, et al. 2026. [Understanding eccentric temperate giants: an in-depth study of the architecture and stellar obliquity of the TOI-2134 system.](#) *MNRAS* **550**, 1265
- [5] Yahalomi, D.A., **Lu, T.**, Armitage, P., Bedell, M., Casey, A.R., Price-Whelan, A. and Rice, M., 2026. [The Astrometric Resoeccentric Degeneracy: Eccentric Single Planets Mimic 2:1 Resonant Planet Pairs in Astrometry.](#) *ApJL* **999**, L9
- [4] Cassese, B., Rice, M. and **Lu, T.**, 2025. [A High-Precision, Differentiable Code for Solar System Ephemerides.](#) *PSJ* **6**, 252
- [3] An, Q., **Lu, T.**, Brandt, G.M., Brandt, T.D. and Li, G., 2025. [Significant Mutual Inclinations Between the Stellar Spin and the Orbits of Both Planets in the HAT-P-11 System.](#) *AJ* **169**, 22
- [2] Cassese, B., Vega, J., **Lu, T.**, Rice, M., Poddar, A. and Kipping, D., 2024. [squishyplanet: Modeling transits of non-spherical planets in JAX.](#) *JOSS* **9**, 100
- [1] Wahl, S., Thorngren, D., **Lu, T.** and Militzer, B., 2021. [Tidal Response and Shape of Hot Jupiters](#) *ApJ* **921**, 105

OTHER CO-AUTHOR

- [4] Naponiello, L., Poretti, E., Rice, K., et al. (inc. **Lu, T.**) 2026. [Refined parameters, formation, and dynamical stability of the wild exoplanet system K2-312=HD 80653.](#) *A&A* in press
- [3] Tajer, H., Wang, J., Childs, A., et al. (inc. **Lu, T.**) 2026. [Origins of Mercury’s Big Heart of Iron: Exploring Pathways to Form High Core Mass Fraction \(CMF\) Planets via N-body Simulations.](#) *ApJ* **998**, 46
- [2] Bardalez Gagliuffi, D.C.†, Balmer, W.O.†, Pueyo, L., et al. (inc. **Lu, T.**) 2025. [JWST Coronagraphic Images of 14 Her c: a Cold Giant Planet in a Dynamically Hot, Multi-planet System.](#) *ApJL* **988**, L18
- [1] Wang, N., Lu, L.Y., Liu, H.G., et al. (inc. **Lu, T.**) 2025. [Development of Gravity Theories in the View of TRAPPIST-1e.](#) *RAA* **25**, 015003

WHITEPAPERS

- [1] Levine, W. G., Gerbig, K., Loudon, E., **Lu, T.**, Hsieh, C., O’Connor, C., Li, R. and Dong, J., 2024. [Emerging Researchers in Exoplanetary Science \(ERES\): Lessons Learned in Conference Organization for Early-Career Researchers.](#) *Bulletin of the AAS* **56**, 2024i004

†: Joint first author

AWARDS & FELLOWSHIPS

- [Yale Dirk Brouwer Prize for Outstanding Thesis of Unusual Merit](#) 2026
- [DDA/AAS Raynor L. Duncombe Student Research Prize](#) 2025
- Flatiron Research Fellowship 2025
- Flipped Science Fair Best Poster Explanation 2024
- AAS International Travel Grant 2024
- Graduate Student Assembly Conference Travel Fellowship 2024
- National Science Foundation GRFP (*Honorable Mention*) 2022
- Joan and Arnold Seidel Griffith Observer Science Writing Contest (*Second Place*) 2022
- Alan Porter Memorial Foundation Fellow 2019
- Summer Undergraduate Research Fellow 2018

RESEARCH MENTORING

- Yurou (Nina) Liu*** Jun 2023 –
Undergraduate Researcher (*Yale*) → Ph.D. student (*Princeton*)
Papers: [Liu, Lu & Rice \(2025\)](#), [Liu, Lu & Rice \(2026\)](#)
- Haedam Im** May 2025 – Apr 2026
Dorrit Hoffleit Undergraduate Research Fellow (*MIT*)
Paper: [Im, Lu, Rice, Tran, Li & Naoz \(2026\)](#)

Eritas Yang
 Ph.D. Student (*Princeton*)
 Paper: [Yang, Lu, Yahalomi & Winn \(2026\)](#)

Vighnesh Nagpal
 Ph.D. Student (*U. Chicago*)

Haniyeh Tajer
 Ph.D. Student (*Ohio State*)
 *Primary Research Advisor

Oct 2025 – Jun 2026

Aug 2025 –

May 2025 –

COLLOQUIA & SEMINARS

Italics: Upcoming

- *Physics & Astronomy Colloquium, SFSU (San Francisco, CA)* Oct 2026
- *Exoplanet Journal Club, University of Chicago (Chicago, IL)* Oct 2026
- *Exoplanet Seminar, Princeton University (Princeton, NJ)* Sep 2026
- *Special Seminar, UCLA (Los Angeles, CA)* Aug 2026
- *Bowler Group Meeting, UCSB (Santa Barbara, CA)* Aug 2026
- *Öberg Group Meeting, Harvard University (Cambridge, MA)* Apr 2026
- *Monday Afternoon Talk, MIT (Cambridge, MA)* Apr 2026
- *Coffee Talk, CCA (New York, NY)* Apr 2026
- *Astronomy & Planetary Science Colloquium, NAU (Flagstaff, AZ)* Feb 2026
- *Astronomy Seminar, AMNH (New York, NY)* Feb 2026
- *Powell Group Meeting, University of Chicago (Chicago, IL)* Oct 2025
- *Tea Talk, Indiana University (Bloomington, IN)* Feb 2025
- *ITC Luncheon, Harvard University (Cambridge, MA)* Jan 2025
- *Exoplanet Pizza Lunch, Harvard University (Cambridge, MA)* Jan 2025
- *Exoplanet Seminar, Princeton University (Princeton, NJ)* Nov 2024
- *Planetary Lunch Seminar, MIT (Cambridge, MA)* Nov 2024
- *Astronomy Seminar, Columbia University (New York, NY)* Oct 2024
- *Astronomy Seminar, Stony Brook University (Stony Brook, NY)* Oct 2024
- *CRA Seminar, Georgia Institute of Technology (Atlanta, GA)* Oct 2024
- *Planetary Science Seminar, Caltech (Pasadena, CA)* Oct 2024
- *Lunch Seminar, UCLA (Los Angeles, CA)* Sep 2024
- *Exoplanet Group Meeting, UCSB (Santa Barbara, CA)* Sep 2024
- *FLASH Talk, UCSC (Santa Cruz, CA)* Sep 2024
- *Star and Planet Lunch, University of Hawai'i (Honolulu, HI)* Aug 2024
- *Makino Lab Seminar, Kobe University (Kobe, Japan)* Jul 2023
- *Exoplanet Seminar, Nanjing University (Nanjing, China)* Jul 2023
- *Exoplanet Seminar, Princeton University (Princeton, NJ)* Mar 2023
- *Astronomy Seminar, Monash University (Monash, Australia)* Feb 2023

CONFERENCE TALKS

Bold: Invited Talk

- CT Exoplanet Picnic, Wesleyan College (*Wesleyan, CT*) Jul 2026
- New York Area Exoplanets Meeting III, AMNH (*New York, NY*) May 2026
- International Conference on Exoplanets and Planet Formation (*Shanghai, China*) Dec 2025
- 246th AAS Meeting Dissertation Talk (*Anchorage, AK*) Jun 2025
- Boston Area Planetary Science Meeting, MIT (*Cambridge, MA*) May 2025
- New York Area Exoplanets Meeting II, CCA (*New York, NY*) May 2025
- 56th DDA Meeting, Georgia Institute of Technology (*Atlanta, GA*) May 2025
- **ML in Planetary Dynamics Workshop**, CCA (*New York, NY*) Apr 2025
- CT Exoplanet Picnic, Wesleyan College (*Wesleyan, CT*) Jul 2024
- Emerging Researchers in Exoplanet Science IX, Cornell University (*Ithaca, NY*) Jul 2024
- **REBOUND 24 Meeting** (*Virtual*) Jul 2024
- New York Area Exoplanets Meeting I, CCA (*New York, NY*) May 2024
- 55th DDA Meeting, University of Toronto (*Toronto, Canada*) May 2024
- 54th DDA Meeting, Michigan State University (*East Lansing, MI*) May 2023
- 240th AAS Meeting (*Pasadena, CA*) Jun 2022

POSTER PRESENTATIONS	▪ Flatiron Scientific Exchange, CCA (<i>New York, NY</i>)	Jun 2026
	▪ 57th DDA Meeting, University of Chicago (<i>Chicago, IL</i>)	Jun 2026
	▪ Emerging Researchers in Exoplanet Science, Princeton University (<i>Princeton, NJ</i>)	Jun 2025
	▪ Exoplanets V Meeting (<i>Leiden, The Netherlands</i>)	Jun 2024
	▪ Extreme Solar Systems (<i>Christchurch, New Zealand</i>)	Jun 2024
	▪ 2022 AGU Meeting (<i>Chicago, IL</i>)	Dec 2022
	▪ Exoplanets IV Meeting (<i>Las Vegas, NV</i>)	May 2022
	▪ 2019 AGU Meeting (<i>San Francisco, CA</i>)	Dec 2019
TEACHING	ASTR 210: Stars and Their Evolution	New Haven, CT, USA
	Yale University Teaching Fellow	Spring 2023
	ASTR 160: Frontiers and Controversies in Astrophysics	New Haven, CT, USA
	Yale University Teaching Fellow	Fall 2022
	ASTR 160: Frontiers and Controversies in Astrophysics	New Haven, CT, USA
	Yale University Teaching Fellow	Spring 2021
OUTREACH	ASTR 155: Introduction to Astronomical Observing	New Haven, CT, USA
	Yale University Teaching Fellow	Fall 2021
	Dean's Tutor	Pasadena, CA, USA
	Tutoring Caltech students in Physics, Astronomy and Computer Science	2017-2020
	Warrior-Scholar Program	New Haven, CT, USA
	Research Project Leader	Summer 2023
	<i>One week intensive boot camp preparing military veterans for undergraduate degrees</i>	
	Yale OpenLabs	
	Local outreach organization dedicated to educating middle school children on STEM topics. Receipt of the Seton Elm-Ivy Award.	
	▪ Public Relations Officer	2023
INFORMAL MENTORING	▪ Regular Outreach Speaker	2022 – 2025
	• Science Cafe 2024 Speaker: <i>What if 'Cotton Candy' Planet are Actually Ring Pops?</i>	
	• Flipped Science Fair Best Poster Explanation: <i>Probing the Shapes of Planets</i>	
	• Science Cafe 2022 Speaker: How We Discover Exoplanets	
	Astronomy on Tap	
	New Haven branch of outreach program aimed at engaging local community with astronomy research	
	▪ Social Media Director	2022
	Science in the News	
	Local outreach organization dedicated to educating the greater community on STEM topics.	
	▪ Outreach Speaker	2021
TELESCOPE PROPOSALS (CO-I)	• Hidden Things: Pulling Back the Veil on the Universe	
	Chris Santiago	2022 – 2024
	Astro Sibb	Yale University
	Kylyn Smith	2023 – 2024
	Astro Sibb	Yale University
	Sally Jiang (current Ph.D. Candidate, Columbia University)	2022 – 2023
	Astro Sibb	Yale University
	James Webb Space Telescope – 44.7 hours	Cycle 5 GO
	ID: 11253. Fluffy or Flashy? Resolving the Mystery of the Coldest-Known Super Puff. <i>PI: Ben Cassese</i>	
	Keck Observatory (KPF) – 0.5 nights	Yale 2026A
	Directly Probing the Migration History of WASP-53 with Precise Orbital Constraints. <i>PI: Quang Tran</i>	
	Keck Observatory (KPF) – 2 nights	Yale 2024B
	The Search for Double Hot Jupiter Systems. <i>PI: Malena Rice</i>	

	Keck Observatory (HIRES) – 2 nights	Yale 2023A
	Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 2). <i>PI: Emma Loudon</i>	
	Keck Observatory (HIRES) – 2 nights	Yale 2022B
	Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 1). <i>PI: Emma Loudon</i>	
PROFESSIONAL SERVICE	Science Organizing Committee – 57th DDA Meeting	June 2026
	<i>Division of Dynamical Astronomy, University of Chicago</i>	
	Science Organizing Committee Chair – DSPS	Jan 2026
	<i>A Dynamical Synthesis of Planetary Systems, New Haven</i>	
	Organizing Committee – ERES X	July 2025
	<i>Emerging Researchers in Exoplanet Science Symposium, Princeton University</i>	
	Science Organizing Committee – ERES VIII	June 2023
	<i>Emerging Researchers in Exoplanet Science Symposium, Yale University</i>	
	Reviewer	
	<i>NASA XRP, NASA Solar System Workings Panel, A&A, AJ, ApJ, MNRAS</i>	2024 – Present
	Main Co-Organizer – Yale Exoplanets and Stars Seminar	2023 – 2025
	<i>Weekly Seminar Series</i>	
	Committee Member – ACDC	2022 – 2025
	<i>Yale University Astronomy Climate and Diversity Committee</i>	
MEDIA	Yale Daily News	Jun 2025
	New study offers a double dose of ‘hot Jupiters’	
	Popular Science	Oct 2024
	What if ‘cotton candy’ planets are actually Ring Pops?	
	AAS Author Journal Series	Nov 2022
	Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9	
COMMUNITY ENGAGEMENT	New Haven Road Runners – New Haven, CT, USA	
	<i>Local USATF sanctioned running club</i>	
	▪ Public Relations Officer	Jul 2023 – Jul 2024
	▪ Racing Team Director / Elite Athlete Coordinator	Jul 2022 – Jun 2025
	NCAA Athlete – Pasadena, CA, USA	2016 – 2020
	<i>NCAA Division III Cross Country/Track & Field at Caltech</i>	
	Avery House – Pasadena, CA, USA	
	<i>Student Government</i>	
	▪ Athletics Manager	2017 – 2018
	▪ Underclassmen Counselor	2017 – 2020

[CV compiled on 2026-08-18]